list of chmod code

The chmod command in Linux is used to change file or directory permissions. Permissions are expressed as **read (r)**, **write (w)**, and **execute (x)**, and they can be set for the owner, group, and others.

Numeric Codes for Permissions

The permissions are represented numerically:

- **4** = Read (r)
- **2** = Write (w)
- **1** = Execute (x)
- **0** = No Permission

Examples of Numeric Permissions

You combine these numbers to set permissions:

Numeric Code	Permission	Explanation
0	---	No permissions
1	--x	Execute only
2	-w-	Write only
3	-wx	Write and execute
4	r--	Read only
5	r-x	Read and execute
6	rw-	Read and write
7	rwx	Read, write, and execute

Common Examples of chmod

When using chmod, you specify the permissions for **owner (u)**, **group (g)**, and **others (o)** in three digits:

1. **chmod 777 file**
    - Full permissions for owner, group, and others.
    - Equivalent: rwxrwxrwx
2. **chmod 755 file**
    - Owner: Full permissions (rwx).

    - Group & others: Read and execute (r-x).
3. **chmod 644 file**
    - Owner: Read and write (rw-).
    - Group & others: Read only (r--).
4. **chmod 600 file**
    - Owner: Read and write (rw-).
    - Group & others: No permissions (---).
5. **chmod 400 file**
    - Owner: Read only (r--).
    - Group & others: No permissions (---).
6. **chmod 000 file**
    - No permissions for anyone.

Using Symbolic Mode

You can also modify permissions symbolically:

- **chmod u+x file**: Add execute permission for the owner.
- **chmod g-w file**: Remove write permission for the group.
- **chmod o+r file**: Add read permission for others.
- **chmod a-r file**: Remove read permission for all (owner, group, others).

Would you like examples specific to files or directories?